Experiment No.: 02

Title: Perform Linear Regression on a data set

Batch: B3 **Roll No.: 16014223083** **Experiment No.: 02**

Aim: To perform Linear Regression analysis using Microsoft Excel in order to study the relationship between an independent variable and a dependent variable and to predict outcomes using the regression model.

Resources needed: VSCode, Python, Google Colab, Jupyter Notebook

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Pre Requisite: Data Science, Machine Learning

Theory:

Linear Regression is a **statistical technique** used to model the relationship between a **dependent variable (Y)** and one or more **independent variables (X)** by fitting a linear equation to observed data.

Simple Linear Regression Model

$Y=\beta 0+\beta 1X$

Where:

- Y = Dependent variable
- X = Independent variable
- $\beta 0$ = Intercept (value of Y when X = 0)
- $\beta 1$ = Slope (change in Y per unit change in X)

Objective of Linear Regression

- To estimate the **best-fit line** using the **least squares method**
- To analyze the **strength of relationship** between variables
- To make **predictions** based on observed data

Assumptions

- Linear relationship between X and Y
- Independence of observations
- Constant variance (homoscedasticity)
- Normally distributed errors

Algorithm / Procedure

Go to **File → Options → Add-ins**.

Select **Excel Add-ins → Go**.

Check **Analysis ToolPak** and click OK.

Go to **Data → Data Analysis → Regression**.

Set:

- **Input Y Range**: Dependent variable data
- **Input X Range**: Independent variable data

Check **Labels** (if headers are included).

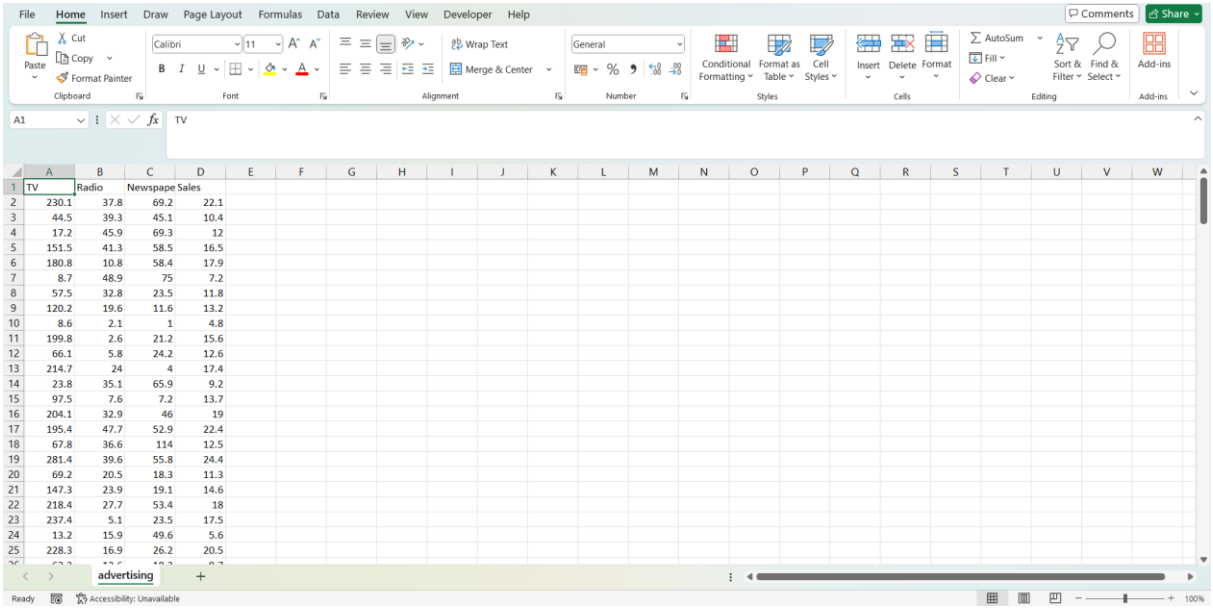
Choose the output location.

Click **OK**.

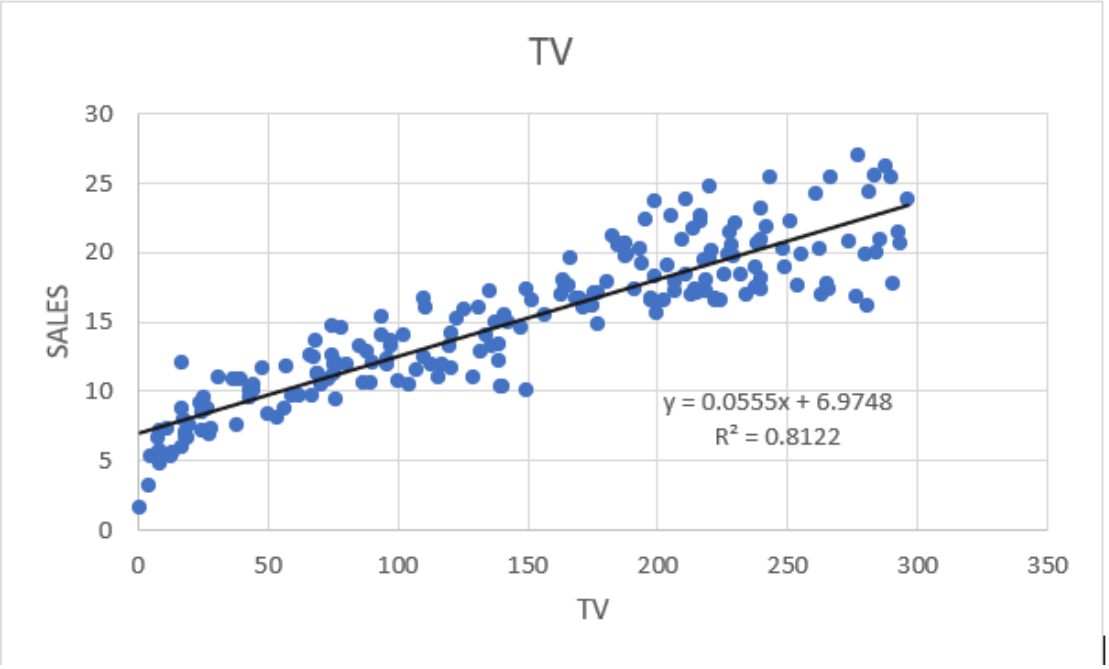
Analyze the output including coefficients, R^2 value, and significance.

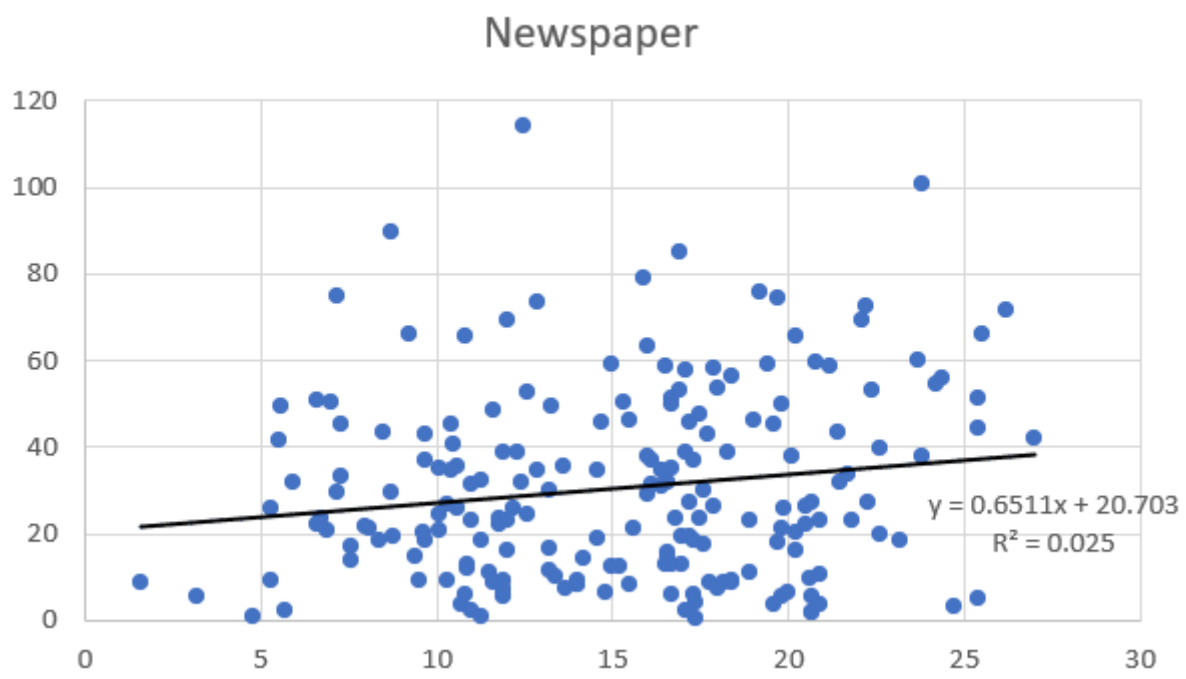
Data Set

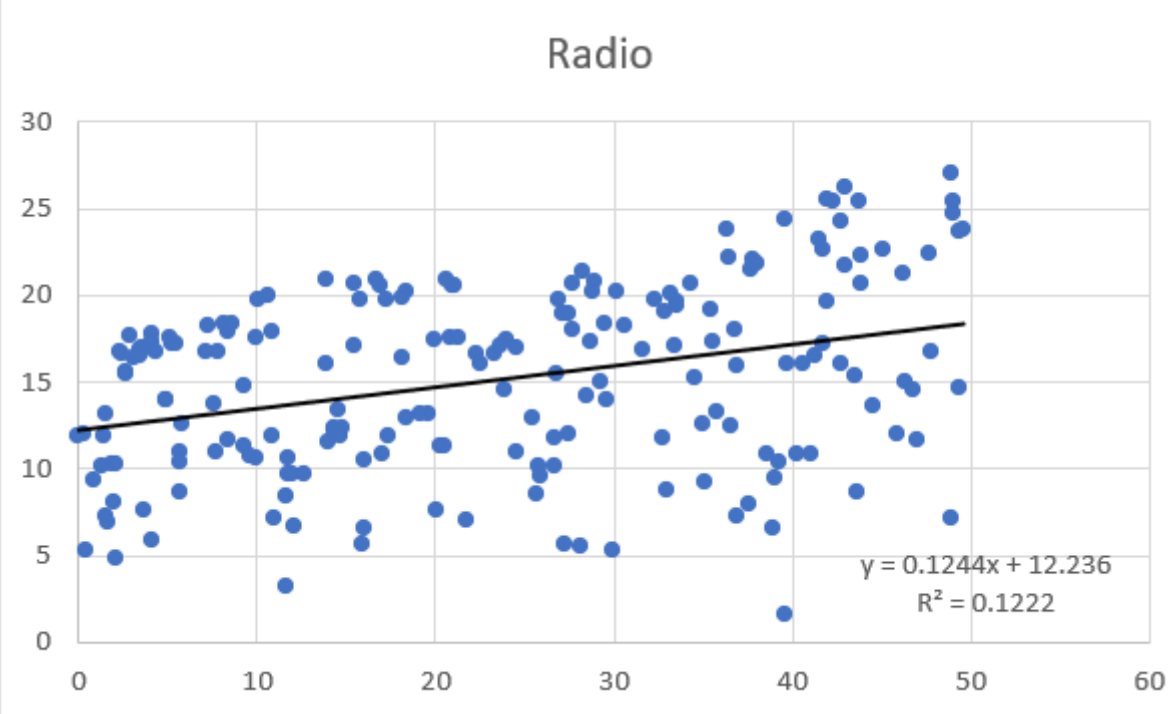
Kindly take the appropriate Data set and mention the reference.
https://www.kaggle.com/code/ashydv/sales-prediction-simple-linear-regression/input



TV	Radio	Newspape	Sales
230.1	37.8	69.2	22.1
44.5	39.3	45.1	10.4
17.2	45.9	69.3	12
151.5	41.3	58.5	16.5
180.8	10.8	58.4	17.9
8.7	48.9	75	7.2
57.5	32.8	23.5	11.8
120.2	19.6	11.6	13.2
8.6	2.1	1	4.8
199.8	2.6	21.2	15.6
66.1	5.8	24.2	12.6
214.7	24	4	17.4
23.8	35.1	65.9	9.2
97.5	7.6	7.2	13.7
204.1	32.9	46	19
195.4	47.7	52.9	22.4
67.8	36.6	114	12.5
281.4	39.6	55.8	24.4
69.2	20.5	18.3	11.3
147.3	23.9	19.1	14.6
218.4	27.7	53.4	18
237.4	5.1	23.5	17.5
13.2	15.9	49.6	5.6
228.3	16.9	26.2	20.5







Outcomes:
CO2: Implement the statistical methods to analyse and interpret the data

Conclusion: (Conclusion to be based on the objectives and outcomes achieved)

In this experiment, we performed Linear Regression analysis using Microsoft Excel to model the relationship between advertising budgets (TV, Radio, Newspaper) and Sales. Using the Ordinary Least Squares (OLS) method, we generated best-fit lines and regression equations for each variable.

TV vs. Sales: This model showed the strongest relationship with a high **R^2 value of 0.8122**, indicating that TV advertising is a reliable predictor of sales.

Radio and Newspaper: These variables showed significantly lower R^2 values (0.1222 and 0.025, respectively), suggesting a weak correlation with sales

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Questions (Viva / Post-Lab)

1. **What is linear regression?**
   Linear regression is a statistical method used to model the relationship between a dependent variable (y) and one or more independent variables (x). It finds the "best-fit" straight line through the data points to predict values or understand relationships. The fundamental equation is: $y = mx + c$
   Where m is the slope and c is the intercept

2. **What does the slope of a regression line indicate?**
   The slope represents the rate of change. It tells you how much the dependent variable (y) is expected to increase or decrease for every one-unit increase in the independent variable (x).
   - Positive slope: As x increases, y increases.
   - Negative slope: As x increases, y decreases.

3. **What is the significance of the R-squared value?**
   The R-squared (R^2) value, or coefficient of determination, measures **goodness of fit**. It represents the percentage of the variance in the dependent variable that can be explained by the independent variable(s).
   - Range: 0 to 1 (0% to 100%).
   - Significance: A higher R^2 generally indicates that the model explains the data well, while a low R^2 suggests the model does not capture the trend effectively.

4. **What are the assumptions of linear regression?**
   For the results to be reliable, four key assumptions must be met:
   - Linearity: The relationship between variables is linear (a straight line).
   - Independence: Observations are independent of each other (no hidden connections).
   - Homoscedasticity: The variance of the errors (residuals) is constant across all levels of x.
   - Normality: The residuals (errors) follow a normal distribution.

5. **How does Excel calculate the regression line?**
   Excel uses the Ordinary Least Squares (OLS) method. This algorithm calculates the line that minimizes the sum of the squared vertical differences (residuals) between the observed data points and the predicted line. You can access this via the LINEST function, the Data Analysis Toolpak, or by adding a trendline to a scatter plot

6. **What is the difference between correlation and regression?**
   Correlation: A single number that measures the strength and direction of the relationship between two variables. It does not distinguish between dependent and independent variables.
   Regression: A mathematical model that defines the nature of the relationship. It identifies a dependent variable and uses it to predict specific outcomes based on the independent variable.

7. **Can linear regression be used for non-linear data? Why or why not?**
   Generally, no. Simple linear regression forces a straight line onto the data. If the data is curved (non-linear), a straight line will result in:
   - Poor predictions (high error rates).
   - Violated assumptions (the residuals will show a pattern rather than being random).

Grade: AA / AB / BB / BC / CC / CD /DD

Signature of faculty in-charge with date

References:

1. Montgomery, D. C., Peck, E. A., & Vining, G. G., *Introduction to Linear Regression Analysis*, Wiley.
2. Draper, N. R., & Smith, H., *Applied Regression Analysis*, Wiley.
3. Microsoft Excel Documentation – Regression Analysis Tool.
4. James, G., Witten, D., Hastie, T., & Tibshirani, R., *An Introduction to Statistical Learning*, Springer.
5. Walpole, R. E., Myers, R. H., Myers, S. L., & Ye, K., *Probability and Statistics for Engineers and Scientists*, Pearson.